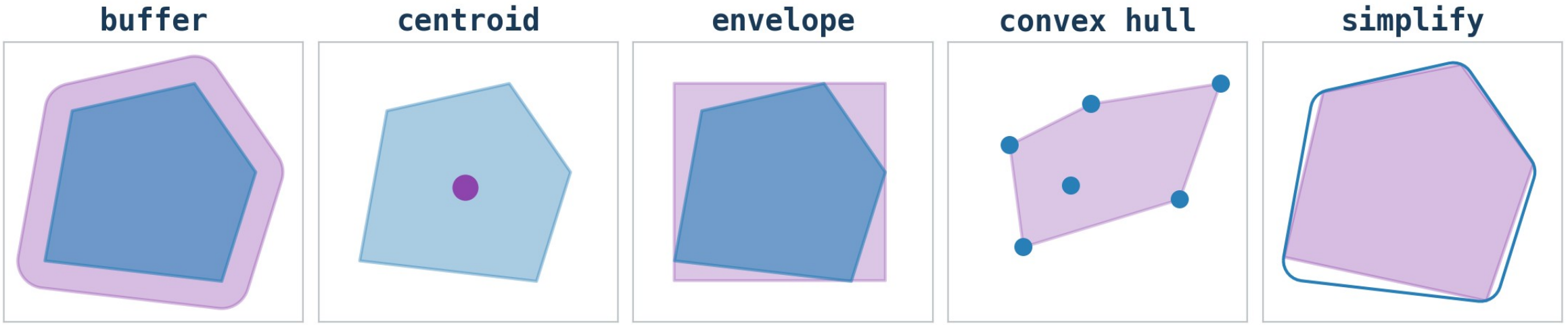
Vector operations

Geometric operations, overlay, buffer, distance and spatial join

Examples: Frohnenbruch, Kamp-Lintfort



One geometry in, one geometry out



In QGIS

Vector → Geoprocessing → Buffer

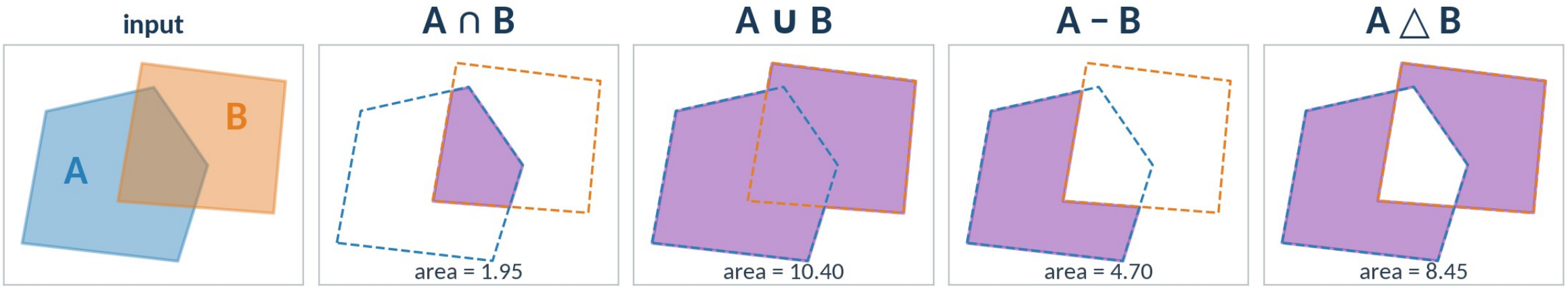
Vector → Geometry Tools → Centroids

Processing → Minimum bounding geometry

Vector → Geometry Tools → Simplify

- Every tool creates a new layer
- Distances in layer units: reproject to EPSG:25832 first

Set operations on two polygons



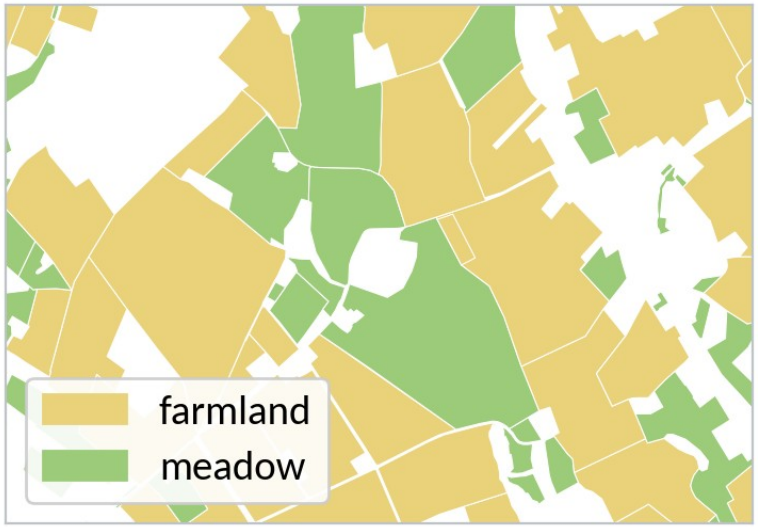
Operation	keeps ...
intersection A ∩ B	area in A and B
union A ∪ B	area in A or B
difference A – B	in A, not in B
sym. difference A △ B	in exactly one

In QGIS

Vector → Geoprocessing Tools →
Intersection, Union, Difference,
Symmetrical Difference
Order matters for Difference

Overlay of two layers: farmland near streams

① farmland + meadow (79 polygons)



② streams, buffer 50 m (dissolved)



③ overlay: 30 new polygons



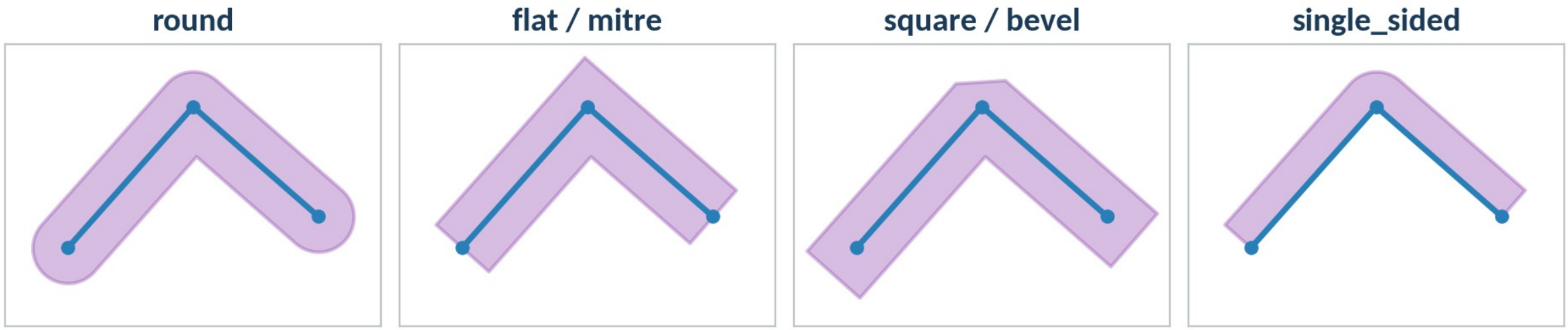
land use	total [ha]	≤ 50 m [ha]	share
farmland	241.4	15.4	6 %
meadow	102.9	18.5	18 %

In QGIS

1 Extract by attribute: farmland, meadow
2 Buffer streams 50 m, dissolve
3 Intersection of 1 and 2
4 Field Calculator: $area / 10000

© OpenStreetMap contributors (ODbL)

Buffer: distance zone around a geometry



In QGIS

Buffer: segments, end cap, join style

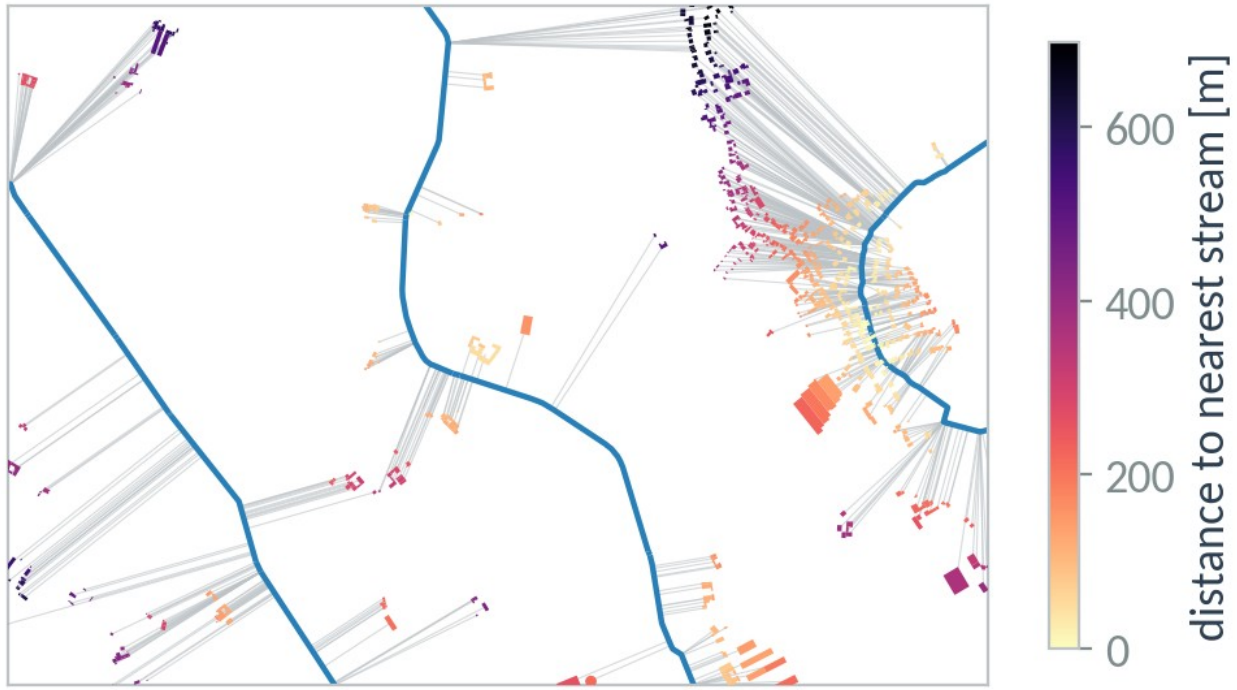
‘Dissolve result’ merges overlaps

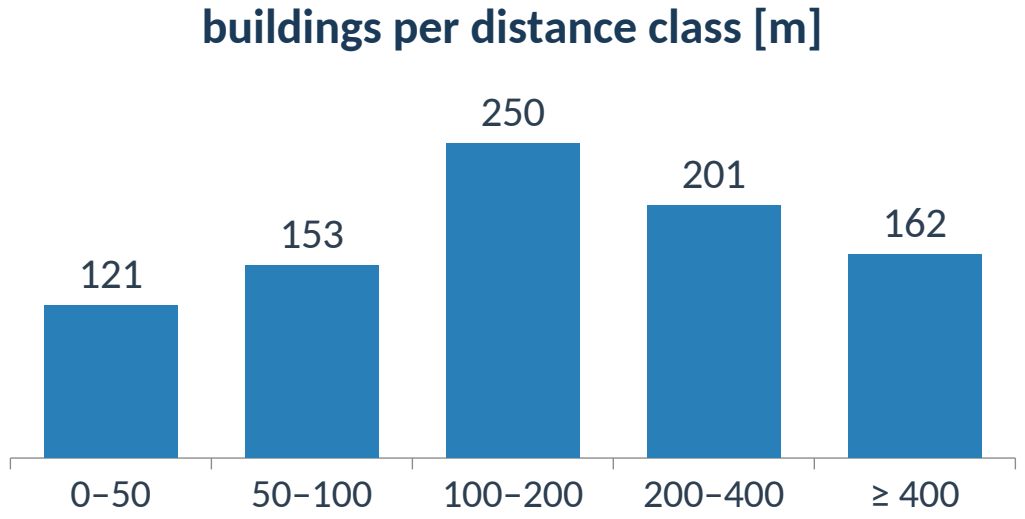
Processing → Single sided buffer

Negative distance shrinks polygons

- Dissolve overlapping buffers before an overlay
- Distance in CRS units: project first

Distance between objects: nearest neighbour





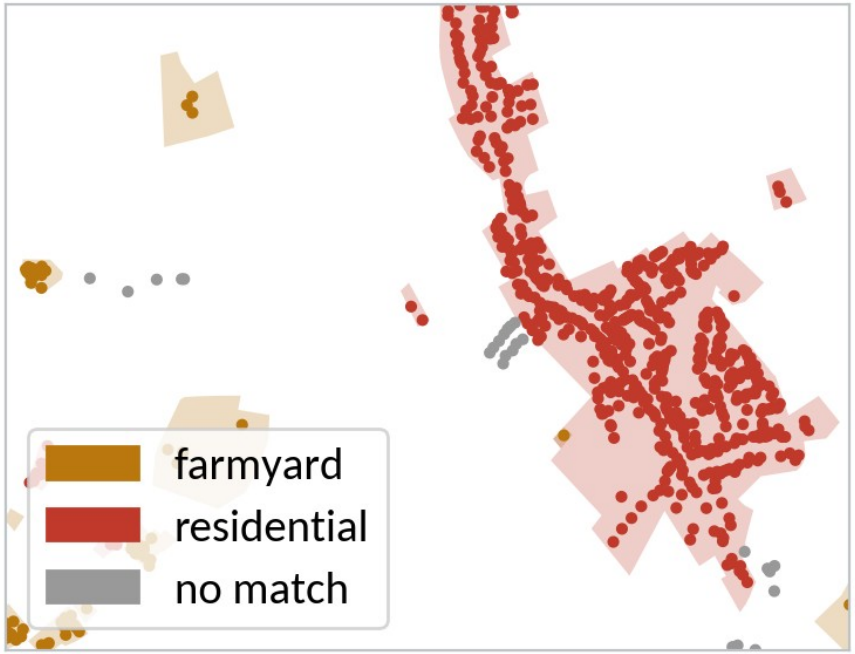
In QGIS

Processing → Join attributes
by nearest (distance field)
Distance to nearest hub: lines
Measure Line: one pair

© OpenStreetMap contributors (ODbL)

Spatial join or overlay?

sjoin: points get the polygon's attributes



overlay: geometries are cut at boundaries



In QGIS

Join attributes by location

Geoprocessing → Intersection

sjoin: attributes travel, geometries stay

overlay: geometries are cut

© OpenStreetMap contributors (ODbL)

Area and length depend on the CRS

CRS	unit	area	perimeter
ETRS89 / UTM 32N (25832)	m	63.2 km²	47.2 km
Web Mercator (3857)	m	162.8 km²	75.8 km
WGS 84 lon/lat (4326)	degree	0.00818 deg²	0.562 deg
ellipsoid (pyproj.Geod)	m	63.2 km²	47.2 km

In QGIS

Field Calculator: $area (ellipsoidal)

area(@geometry): planar, layer CRS units

Vector → Data Management → Reproject Layer

× 2.58

Web Mercator area vs. UTM

Measure in a projected CRS that suits the area, never in lon/lat degrees.

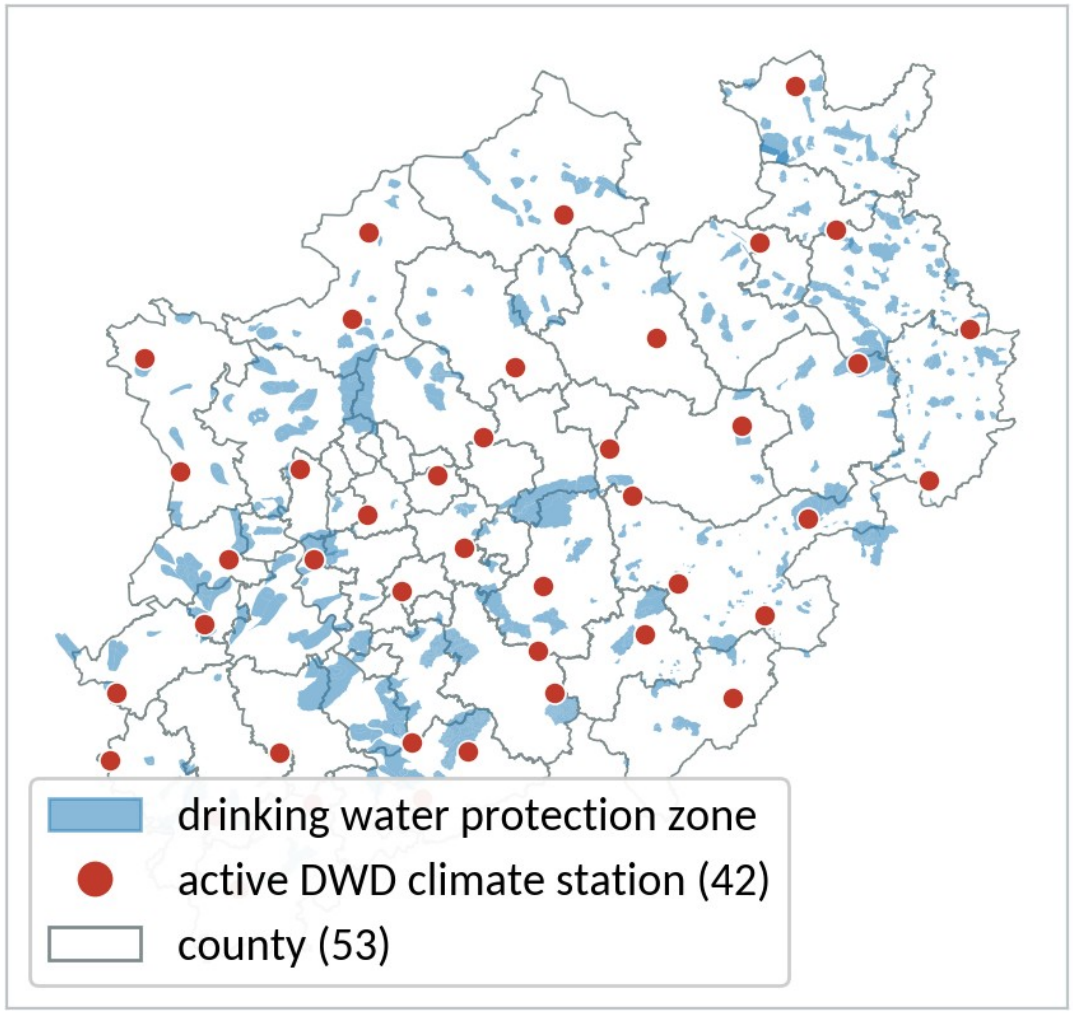


The same operations in three tools

Operation	shapely / GeoPandas	QGIS Processing	PostGIS
buffer	gdf.buffer(d)	Buffer	ST_Buffer(geom, d)
intersection overlay	gpd.overlay(a, b, how='intersection')	Intersection	ST_Intersection(a.geom, b.geom)
union / dissolve	gdf.dissolve(by=…), union_all()	Dissolve, Union	ST_Union(geom) … GROUP BY
difference	gpd.overlay(a, b, how='difference')	Difference	ST_Difference(a.geom, b.geom)
spatial join	gpd.sjoin(a, b, predicate='within')	Join attributes by location	JOIN … ON ST_Within(…)
nearest	gpd.sjoin_nearest(a, b)	Join attributes by nearest	ORDER BY a.geom <-> b.geom LIMIT 1
distance	a.distance(b)	Distance matrix	ST_Distance(a.geom, b.geom)
area / length	gdf.area, gdf.length	Field calculator $area	ST_Area, ST_Length



Practice: stations and protection zones



Q1 How well do the climate stations cover each county?

Q2 How much of each county is drinking water protection zone?

In QGIS

Q1 Reproject layer → EPSG:25832
Count points in polygon
Buffer 20 km, Dissolve result
Difference (county – cover)

Q2 Extract by expression (F, T)
Dissolve
Intersection (county ∩ zones)
Field Calculator: area(@geometry)

Stations: Deutscher Wetterdienst (CC BY 4.0) · counties (DVG1): © Geobasis NRW, dl-de/zero-2-0 · protection zones: Land NRW, opengeodata.nrw.de, dl-de/zero-2-0

Vector data from a web service: WFS



	WMS	WFS
you get	a picture	features + attributes
buffer, join, area	no	yes
example (NRW)	DTK, DOP	counties, field parcels

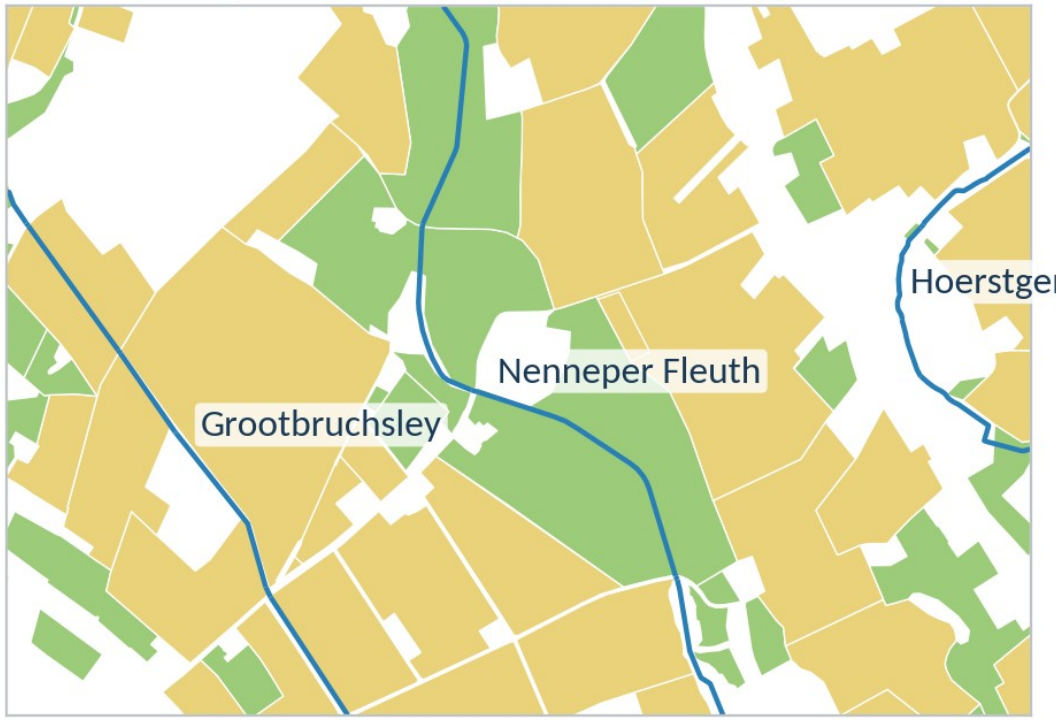
In QGIS

Layer → Data Source Manager →
WFS / OGC API - Features → New
URL: https://www.wfs.nrw.de/umwelt/lwk_eufoerderung → Connect
Zoom in first: only the view is sent

Field parcels, strip search area: Landwirtschaftskammer NRW, dl-de/by-2-0 · © OpenStreetMap contributors (ODbL)

OpenStreetMap or official data?

OpenStreetMap landuse: 343 ha



declared field parcels (LWK): 317 ha



OSM: worldwide, volunteered, no ditches here

Official: crop per year, NRW only, dl-de/by

Which one would you use for a fertiliser rule along streams, and which one in your home country?

© OpenStreetMap contributors (ODbL) · Field parcels, strip search area: Landwirtschaftskammer NRW, dl-de/by-2-0

Sources and further reading

- **Data:** © OpenStreetMap contributors (ODbL); municipality boundary from OSM via Nominatim; Stations: Deutscher Wetterdienst (CC BY 4.0) · counties (DVG1): © Geobasis NRW, dl-de/zero-2-0 · protection zones: Land NRW, opengeodata.nrw.de, dl-de/zero-2-0; Field parcels, strip search area: Landwirtschaftskammer NRW, dl-de/by-2-0 (WFS https://www.wfs.nrw.de/umwelt/lwk_eufoerderung)
- **Libraries:** shapely 2 (GEOS), GeoPandas (overlay, sjoin, sjoin_nearest), pyproj
- **Further reading (links):** GeoPandas user guide – Set operations with overlay, Spatial joins; Shapely user manual; Automating GIS Processes (Univ. of Helsinki) – vector analysis lessons; QGIS Training Manual – Vector analysis; Lovelace, Nowosad, Muenchow: Geocomputation with Python, ch. 3–4 (py.geocompx.org, open book); OGC Web Feature Service standard (ogc.org/standards/wfs)
- **Figures:** own work, generated by make_slides.py in this unit

